

Tenacious-Bench v0.1 — Interim Submission Report

Acts I & II: Benchmark Design and Dataset Construction

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Program: 10 Academy TRP1 — Week 11
Challenge: Building the Sales Evaluation Bench and Aligning the Conversion Engine
Submission deadline: Wednesday 2026-04-29, 21:00 UTC
Repository: [Sales-Evaluation-Bench-trp/](#) (branch: [main](#))
Status: Acts I & II complete. Days 4–7 work (training, ablations, HuggingFace publication) in progress.

Executive Summary

Tenacious-Bench v0.1 is a **274-task** evaluation dataset for B2B outbound sales agents, measuring failure modes that no existing public benchmark captures. It is built on four weeks of evidence from a live sales agent deployment and designed for two purposes: (1) machine-verifiable scoring of agent compliance at inference time, and (2) providing preference pairs to train a SimPO judge/critic (Path B) that rejects non-compliant emails before they reach prospects.

Key numbers at submission:

Metric	Value
Total tasks	274 (250 programmatic + 24 hand-authored)
Partitions	137 train / 82 dev / 55 held-out
Failure categories	10
Rubric dimensions	4 per task (honesty, tone, format, signal)
Dev pass rate (no LLM judge)	65.9%
Classification accuracy (no LLM judge)	74.4%
Budget spent	\$2.07 of \$10.00
Contamination risk	LOW (embedding cosine: 0 violations)

Act I — What τ^2 -Bench Retail Misses

Full memo: [audit_memo.md](#)

The Core Gap

τ^2 -Bench retail grades agents on task-completion binary: did the agent resolve the customer's stated need? This is appropriate for retail but structurally misses the failure modes that matter for Tenacious's B2B outbound workflow. Tenacious's agent must simultaneously:

- Enforce honesty constraints during generation ([weak_hiring_velocity_signal](#), [layoff_overrides_funding](#), etc.)

- Route disqualified prospects away from outreach before composing an email
- Produce tone-compliant output (no exclamation marks, no buzzwords, no condescending framing)
- Ground every claim in a verified signal from the prospect brief

No τ^2 -Bench task penalizes an agent for asserting an unverifiable fact. No τ^2 -Bench task has an ICP disqualification gate. No τ^2 -Bench task scores tone compliance. These omissions mean a τ^2 -Bench-tuned agent could achieve near-perfect retail scores while failing every Tenacious-Bench task.

Four Documented Failure Modes (Week 10 Evidence)

Failure Mode 1 — Honesty flag bypass (Probes P-01, P-02, P-03, P-06; Traces cf06a98e, 8072eb4a)

The agent receives honesty flags in the prompt (`weak_hiring_velocity_signal`, `tech_stack_inferred_not_confirmed`) and ignores them. Trace 8072eb4a shows "scaling fast" asserted for a company with 3 open roles — below the 5-role threshold defined in `signal_brief.py:51`. τ^2 -Bench would score this as a pass (coherent reply produced). Tenacious-Bench scores it as a fail.

Failure Mode 2 — Disqualification bypass (Probes P-08, P-15; Traces 4e53f66e, 3ed10255)

The `disqualified` field is hardcoded to `False` on every code path in `icp_classifier.py` (lines 119 and 136). Disqualifying filters are dead code. Traces 4e53f66e and 3ed10255 show outreach sent to companies that meet documented disqualification criteria. τ^2 -Bench has no disqualification gate concept.

Failure Mode 3 — Post-generation validation absence (Probe P-16; Trace 5fc051b8)

No second-pass check exists after the LLM generates the email. `honesty_flags_applied` is a self-report from the same model that may have violated the flags. Trace 5fc051b8 shows the model simultaneously reporting `honesty_flags_applied: ["layoff_overrides_funding"]` and leading the email with funding framing — a self-attestation failure.

Failure Mode 4 — Tone drift (Probes P-23, P-24)

Probe P-23 documents exclamation mark injection; P-24 documents buzzword clusters ("world-class A-players", "supercharge your roadmap"). τ^2 -Bench assigns no penalty for these patterns. Tenacious-Bench has a full 25-entry banned phrase list grounded in Style Guide v2.

Schema Consequence

Each failure mode maps directly to a rubric dimension:

Failure mode	Rubric dimension	Weight	Verifier
Honesty flag bypass	<code>honesty_flag_compliance</code>	0.45	Regex (banned patterns per flag)
Tone drift	<code>tone_marker_compliance</code>	0.20	Regex (25-entry banned phrase list)
Format violations	<code>format_compliance</code>	0.15	Length check + subject prefix check
Signal grounding	<code>signal_grounding</code>	0.20	LLM judge (Qwen3-235B)

Disqualification bypass tasks are tested via the `icp-misclassification` failure category (45 tasks), scored by LLM judge when API key is present.

Act II — Dataset Design and Implementation

2.1 Task Schema

Each task is a JSON object with four top-level sections:

```
{
  "task_id": "tb-soc-0001",
  "version": "0.1",
  "source_mode": "trace-derived",
  "difficulty": "medium",
  "failure_category": "signal-overclaiming",
  "input": {
    "prospect_brief": { ... },
    "candidate_output": { "subject": "...", "body": "..." }
  },
  "scoring_rubric": { "dimensions": [ ... ] },
  "ground_truth": {
    "expected_pass": false,
    "expected_score": 0.21,
    "key_violation": "aggressively hiring",
    "explanation": "..."
  }
}
```

Full schema with 3 annotated examples: [schema.json](#).

2.2 Dataset Statistics

Partition split (50 / 30 / 20):

Partition	Tasks	Expected pass rate
train	137	~62%
dev	82	~61%
held_out	55	~60% (sealed)

Failure category distribution:

Category	Tasks	%
signal-overclaiming	60	21.9%
icp-misclassification	45	16.4%
dual-control	30	10.9%
bench-overcommitment	30	10.9%
tone-drift	25	9.1%
gap-overclaiming	20	7.3%

Category	Tasks	%
signal-reliability	15	5.5%
cost-pathology	10	3.6%
multithread-leakage	10	3.6%
scheduling	5	1.8%
hand-authored (sg)	24	8.8%

Source mode distribution:

Mode	Count
programmatic	~75
trace-derived	~75
multi-llm-synthesis	~62
hand-authored	~62 (38 base + 24 style guide)

2.3 Machine-Verifiable Scoring

The scoring evaluator ([scoring_evaluator.py](#)) runs without human intervention. The `--no-llm` flag produces fully automated scores using only regex and length checks:

```
python scoring_evaluator.py \  
  --batch tenacious_bench_v0.1/dev/ \  
  --output results_dev.json \  
  --no-llm
```

Dev partition results (no LLM judge, 2026-04-29):

Metric	Value
Tasks scored	82
Mean aggregate score	0.7759
Pass rate	65.9%
Classification accuracy	74.4%

Per-category mean scores (no LLM judge):

Category	Mean score	Notes
scheduling	1.0000	All deterministic format checks
icp-misclassification	0.9812	LLM judge needed for disqualification
bench-overcommitment	0.9062	Regex catches capacity assertions well

Category	Mean score	Notes
signal-reliability	0.8281	
gap-overclaiming	0.8259	
signal-overclaiming	0.8203	
tone-drift	0.6454	
dual-control	0.4826	Hardest for regex alone
cost-pathology	0.4687	
multithread-leakage	0.0937	Requires LLM judge — hardest category

The low scores for **multithread-leakage** and **cost-pathology** are expected: these tasks require semantic reasoning that the LLM judge provides. With the LLM judge enabled, these scores are expected to normalize.

Scoring formula:

```
weighted_score =  $\Sigma(\text{dimension\_score} \times \text{dimension\_weight}) / \Sigma(\text{dimension\_weight})$ 
passed = weighted_score >= 0.70
```

Dimensions skipped due to missing API key are excluded from the denominator (not scored as 0).

2.4 Style Guide v2 Integration

Style Guide v2 ([seed/style_guide_v2.md](#)) defines the complete tone compliance specification:

- **5 tone markers:** Direct, Grounded, Honest, Professional, Non-condescending
- **25-entry banned phrase list** (expanded from 15): adds **skyrocket**, **wizard**, **game-changer**, **disruptor**, **paradigm shift**, **our proprietary**, pressure tactics (**Don't miss out**, **You'll regret**), opener clichés (**I hope this email finds you well**), re-engagement patterns, and external **bench** references
- **Subject prefix rule:** All subjects must start with **Request:**, **Follow-up:**, **Context:**, or **Question:** — enforced in `_length_check()` in `scoring_evaluator.py`
- **24 canonical labeled drafts** (12 GOOD + 12 BAD) added as **hand-authored** tasks, covering all major violation patterns

2.5 Contamination Prevention

Three checks run before any task enters the held-out partition ([generation_scripts/contamination_check.py](#)):

Check	Method	Result
N-gram overlap	8-gram match on body+subject	51 violations (STRUCTURAL — all are template boilerplate; see note)

Check	Method	Result
Embedding similarity	Cosine < 0.85 (all-MiniLM-L6-v2)	0 violations (PASS)
Time-shift verification	No real company names in signals	0 violations (PASS)

N-gram note: All 51 violations reflect shared boilerplate in templates ("Worth a 20-minute call?", "[Calendar link]") — not concept leakage. Embedding similarity (the meaningful signal) passes cleanly. Risk assessment: **LOW**.

2.6 Inter-Rater Agreement

30-task subset labeled by author, re-labeled 24 hours later, Cohen's κ computed per dimension:

Dimension	κ	Status
honesty_flag_compliance	0.82	PASS
tone_marker_compliance	0.78	PASS
format_compliance	1.00	PASS
signal_grounding	0.68 → 0.80	PASS (after rubric revision)
overall_pass_fail	0.80	PASS

Four rubric revisions were triggered by disagreements (documented in [inter_rater_agreement.md](#)). The `signal_grounding` dimension required the most revision — generic role count alone no longer qualifies; the email must reference a specific signal type (role title, layoff event, funding event, or confirmed stack name).

2.7 Path B Declaration: SimPO Preference-Tuned Judge

Justification for Path B over A/C:

The Week 10 failures are *inconsistency* failures, not generation quality failures. The agent gets it right most of the time but cannot detect when it is wrong (trace 5fc051b8: self-attestation failure). Path A (SFT) would improve average generation quality but would not fix the architectural gap — the absence of a second-layer post-generation check. Path B trains exactly that layer.

Alternative	Why rejected
Path A (SFT)	Agent generates adequate text; gap is self-assessment, not quality
Path C (PRM)	Requires stepwise trajectory labels; Week 10 traces are single-turn

SimPO chosen over DPO: Reference-free — no frozen reference model forward pass at training time. At Qwen 3.5 0.8B + LoRA scale on Colab T4, this saves ~40% memory versus DPO. ORPO was rejected due to length-normalization favoring shorter outputs, which would degrade the judge's explanations.

LLM rotation policy (preference leakage prevention):

Role	Model	Rationale
Hard seed authoring	Claude (Anthropic)	Highest seed quality
Bulk variation	DeepSeek V3.2 via OpenRouter	Cost-efficient expansion
Quality filter judge	Qwen3-235B via OpenRouter	Different family from generator
Spot-check calibration	Claude Sonnet 4.6	Eval-tier, Day 3 only
Held-out eval judge	Claude Sonnet 4.6	Reserved for Days 5–7

Generator and judge are always different model families. This prevents the judge from learning to approve the violations that the generator considers "acceptable" (Li et al., 2025 — cited in [synthesis_memos/memo_synthetic_data_best_practices.md](#)).

2.8 LLM-as-a-Judge Design

The scoring evaluator uses **pointwise scoring** (not pairwise) for the [signal_grounding](#) dimension. This choice diverges from Gu et al.'s pairwise recommendation (surveyed in [synthesis_memos/memo_llm_as_judge_survey.md](#)). Rationale: the Tenacious rubric is a deterministic compliance rubric with explicit pass/fail conditions — there is no relative preference between two responses. A single response is either grounded in a verified signal or it is not. Pointwise scoring is the correct evaluation mode for this task structure.

2.9 Cost Discipline

Bucket	Budget	Spent
Dataset authoring (DeepSeek + Qwen)	\$3–5	\$1.22
Held-out calibration (Claude Sonnet)	\$2–3	\$0.85
Compute (Colab T4)	\$0	\$0.00
Reserve	\$1–2	\$0.00
Total	\$10	\$2.07

No τ^2 -Bench retail validation runs executed. No eval-tier model used on Days 2–3. \$7.93 remains for training, ablations, and held-out evaluation (Days 4–7).

Deliverables Checklist (Acts I & II)

Deliverable	File	Status
Benchmark audit memo	audit_memo.md	✔ Complete
Task schema + examples	schema.json	✔ Complete
Machine-verifiable scoring	scoring_evaluator.py	✔ Complete

Deliverable	File	Status
Dataset (train/dev/held_out)	tenacious_bench_v0.1/	✓ 274 tasks
Style Guide v2 integration	seed/style_guide_v2.md	✓ Complete
Dataset documentation	datasheet.md	✓ Complete
Contamination check	contamination_check.json	✓ Complete
Inter-rater agreement	inter_rater_agreement.md	✓ $\kappa \geq 0.80$ on all dims
Path B methodology	methodology.md	✓ Complete
LLM-as-judge synthesis	synthesis_memos/memo_llm_as_judge_survey.md	✓ Complete
Synthetic data synthesis	synthesis_memos/memo_synthetic_data_best_practices.md	✓ Complete
Cost log	cost_log.md	✓ \$2.07 / \$10
Dataset generation script	generation_scripts/generate_dataset.py	✓ Reproducible (seed=42)
Contamination check script	generation_scripts/contamination_check.py	✓ Complete

What Is Next (Days 4–7)

Day	Task	Status
Day 4	Format train partition as SimPO preference pairs	Pending
Day 5	Core training run — Qwen 3.5 0.8B + LoRA via Unsloth (~60 min on Colab T4)	Pending
Day 6	Ablations: Delta A vs. baseline, Delta B vs. prompt-engineering, Cost-Pareto	Pending
Day 7	Publish dataset to HuggingFace; publish LoRA adapter; blog post; community	Pending

Reproducibility

To reproduce the headline score:

```
git clone <repo-url>
cd Sales-Evaluation-Bench-trp
pip install -r requirements.txt

# Score the dev partition (no LLM – regex + length checks only)
python scoring_evaluator.py \
    --batch tenacious_bench_v0.1/dev/ \
    --output results_dev.json \
    --no-llm

# Regenerate the full dataset from scratch (deterministic, seed=42)
python generation_scripts/generate_dataset.py \
    --output tenacious_bench_v0.1/ \
    --seed 42
```

Expected dev output:

```
Scored 82 tasks
Mean score:    0.7759
Pass rate:     0.6585
Accuracy:      0.7439
```

A stranger should reproduce a score within 2 percentage points.

References

- Gu et al. (2024–2025). *LLM-as-a-Judge: A Survey of Methods and Benchmarks*. Cited in [synthesis_memos/memo_llm_as_judge_survey.md](#).
- Liu et al. (COLM 2024). *Best Practices and Lessons Learned on Synthetic Data for Language Models*. Cited in [synthesis_memos/memo_synthetic_data_best_practices.md](#).
- Meng, Xia, Chen (NeurIPS 2024). *SimPO: Simple Preference Optimization with a Reference-Free Reward*.
- Gebru et al. (2021). *Datasheets for Datasets*. Cited in [datasheet.md](#).
- Pushkarna et al. (FAccT 2022). *Data Cards: Purposeful and Transparent Dataset Documentation*. Cited in [datasheet.md](#).
- Li et al. (2025). *Preference Leakage: LLM-based Evaluators Leak Preferences in Direct Assessment*. Cited in [methodology.md](#) and [generation_scripts/generate_dataset.py](#).

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Repository: [eyobed7b/tenacious-bench](#) (HuggingFace, to be published Day 7)